

A ghost mechanism: An analytical model of abrupt learning

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Abrupt learning is commonly observed in neural networks, where long plateaus in network performance are followed by rapid convergence to a desirable solution. Yet, despite its common occurrence, the complex interplay of task, network architecture, and learning rule has made it difficult to understand the underlying mechanisms. Here, we introduce a minimal dynamical system trained on a delayed-activation task and demonstrate analytically how even a one-dimensional system can exhibit abrupt learning through ghost points rather than bifurcations. Through our toy model, we show that the emergence of a ghost point destabilizes learning dynamics. We identify a critical learning rate that prevents learning through two distinct loss landscape features: a no-learning zone and an oscillatory minimum. Testing these predictions in recurrent neural networks (RNNs), we confirm that ghost points precede abrupt learning and accompany the destabilization of learning. We demonstrate two complementary remedies: lowering the model output confidence prevents the network from getting stuck in no-learning zones, while increasing trainable ranks beyond task requirements (*i.e.*, adding sloppy parameters) provides more stable learning trajectories. Our model reveals a bifurcation-free mechanism for abrupt learning and illustrates the importance of both deliberate uncertainty and redundancy in stabilizing learning dynamics.

Understanding the mathematical principles that govern learning dynamics is crucial for advancing the frontiers in both natural and artificial neural networks [1–8]. Empirically, networks often learn to internalize the structure of tasks and exhibit sudden drops in their loss function, which mark the acquisition of new computational capabilities [9–19]. Such abrupt transitions frequently accompany instabilities in learning dynamics, including exploding or vanishing gradients and abrupt shifts in network behavior [20–22]. While a variety of prior works exhibits slow learning followed by sudden performance jumps [23, 24], the precise mechanisms underlying this phenomenon remain poorly understood.

One recent productive approach in this direction involves using dynamical systems theory to study RNNs through the lens of dynamical systems theory [25–32]. During training, RNNs often show destabilized gradients, leading to well-documented learning instabilities [9, 20, 21, 33, 34]. Analysis of RNNs’ attractor landscapes [9, 11, 25, 34–36] has identified bifurcations (*i.e.*, shifts in system behavior, which alter the number or stability of fixed points, periodic orbits, or other invariant sets [33, 37–43]) as key events facilitating the acquisition of novel capabilities [9, 11]. Traditional work in dynamical system theory has established several mechanisms that support robust system behavior without originating from bifurcations. For instance, ghost points, where state variables change very slowly (unlike fixed-points where no change occurs) [37], can help track time. Therefore, limiting the analysis of learning dynamics to bifurcations may overlook important processes that drive abrupt learning.

We start this letter by illustrating the sudden learning phenomenon using a simple example with recurrent neural networks (RNNs). In this “*delayed-activation task*,” networks are initialized to a specific state and must output zero for a delay period, T , after which the output is expected to “activate” and become one. As discussed in **End Matter**, (low-rank [28]) RNNs trained on this task exhibit abrupt learning, where long plateaus in loss value are followed by rapid convergence. To study this phenomenon systematically, we introduce an analytically tractable one-dimensional dynamical system trained on the delayed-activation task and demonstrate how this system exhibits abrupt learning despite its simplicity. Through analysis of this system, we identify a novel mechanism underlying abrupt learning without bifurcations, characterized by the rapid emergence of ghost points (approximate saddle points [37]). Next, we demonstrate that RNNs’ learning dynamics can be substantially stabilized by artificially reducing the confidence level of their outputs and increasing the number of trainable ranks. The former observation underscores the importance of secondary, not goal-directed, mechanisms to stabilize learning, whereas the latter has implications for learning dynamics in low-rank RNNs [28–32]: The presence of “sloppy” (*e.g.*, randomly connected) ranks, which may not necessarily be part of the computation in the fully trained (effectively low-rank) RNNs [29, 32], may support stable learning dynamics.

Our toy model (Fig. 1) is a one-dimensional dynamical system that follows the prototypical form of the saddle-node bifurcation ($\dot{x} = x^2 + r$, [37]). Here, $x \in \mathbb{R}$ is the scalar state variable and $r \in \mathbb{R}$ is the *learnable* parameter.